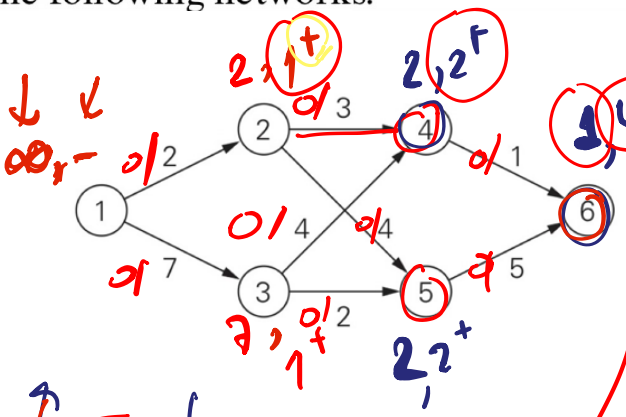
2. Apply the shortest-augmenting path algorithm to find a maximum flow and a minimum cut in the following networks.

b.



u_{ij}: edge capacity

x_{ij}: used amount

r_{ij}: residual amount

l_j: current flow on the vertex j

forward

$l_j = \min\{l_i, r_{ij}\}$

$l_j = \min\{l_i, x_{ij}\}$

backward edge

$r_{ij} = u_{ij} - x_{ij}$

θ: 1 2 3 4 5 6

$l_2 = \min\{l_1, r_{12}\} = 2$ ($l_1 = \infty$, $r_{12} = 2$)

$l_4 = \min\{l_2, r_{24}\} = 2$ ($l_2 = 2$, $r_{24} = 3$)

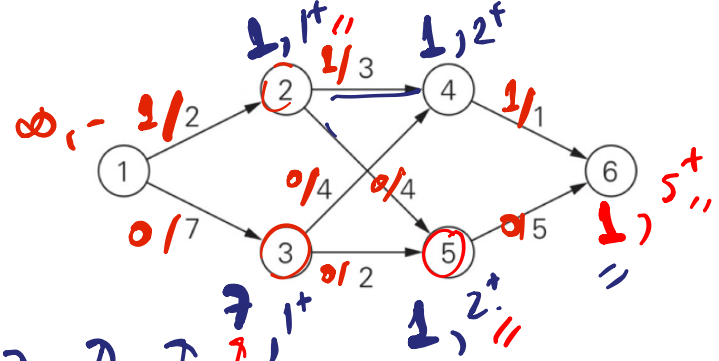
$l_5 = \min\{l_2, r_{25}\} = 2$ ($l_2 = 2$, $r_{25} = 4$)

$l_6 = \min\{l_4, r_{46}\} = 1$ ($l_4 = 2$, $r_{46} = 1$)

Augmenting path: $1 \to 2 \to 4 \to 6$

flow: 1

b.



θ: 1 2 3 4 5

$l_2 = \min\{l_1, r_{12}\} = 1$ ($l_1 = \infty$, $r_{12} = 2-1 = 1$)

$l_3 = \min\{l_1, r_{13}\} = 7$

$l_4 = \min\{l_2, r_{24}\} = 1$ ($l_2 = 1$, $r_{24} = 2$)

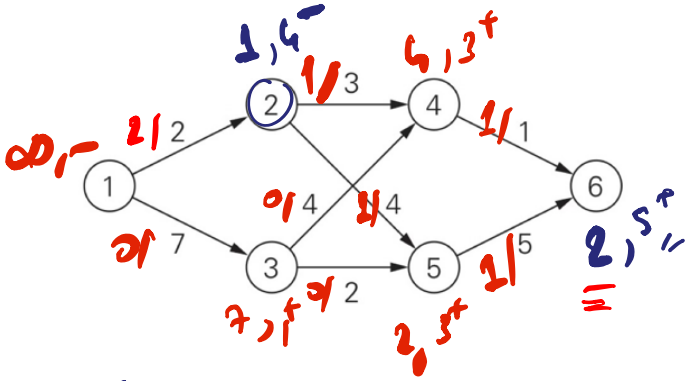
$l_5 = \min\{l_2, r_{25}\} = 1$ ($l_2 = 1$, $r_{25} = 4$)

$l_6 = \min\{l_5, r_{56}\} = 1$ ($l_5 = 1$, $r_{56} = 5$)

path: $1 \to 2 \to 5 \to 6$

flow: 1

b.



$l_4 = \min\{l_3, r_{34}\}$
$\quad\quad 7 \quad\quad 4$

$l_5 = \min\{l_3, r_{35}\} = 2$
$\quad\quad 7 \quad\quad 2$

$l_2 = \min\{l_4, x_{42}\}$
$\quad\quad 4 \quad\quad 1$

$l_6 = \min\{l_5, r_{56}\}$
$\quad\quad 11 \quad\quad 11$
$\quad\quad 2 \quad\quad 4$

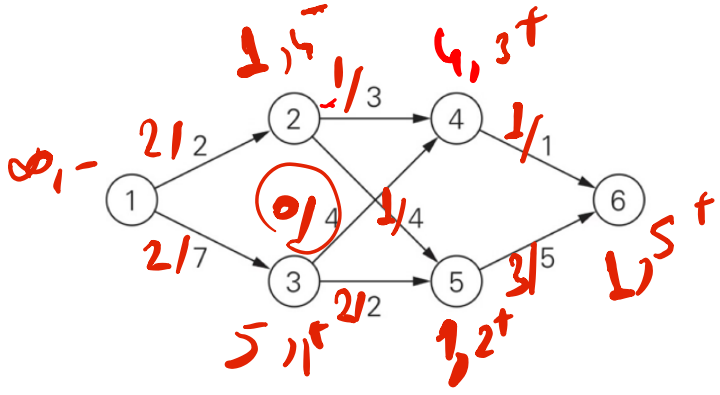
Q: 1 3 4 5 2

Path: $1 \to 3 \to 5 \to 6$

Flow: 2

---

b.



$l_3 = \min\{l_1, r_{13}\}$
$\quad\quad 5 \quad\quad 5$

$l_4 = \min\{l_3, r_{34}\}$
$\quad\quad 5 \quad\quad 4$

$l_2 = \min\{l_4, x_{42}\}$
$\quad\quad 4 \quad\quad 1$

$l_5 = \min\{l_2, r_{25}\}$
$\quad\quad 2 \quad\quad 3$

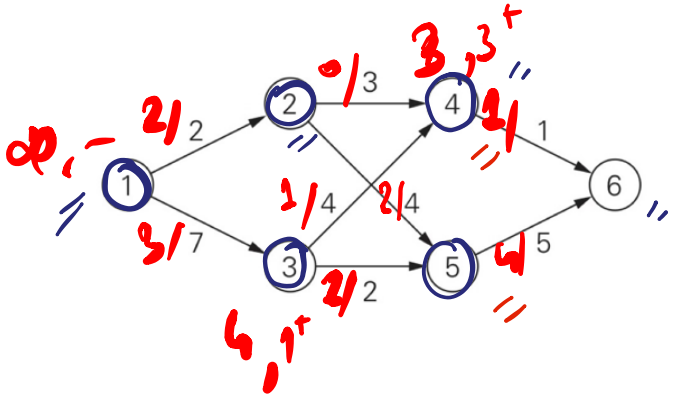
$l_6 = \min\{l_5, r_{56}\}$

Q: 1 3 4 2 5

Path: $1 \to 3 \to 4 \leftarrow 2 \to 5 \to 6$

Flow: 1

b.



$l_4 = \min\{\frac{l_3}{4}, \frac{r_{34}}{3}\}$

θ: 1 3 4

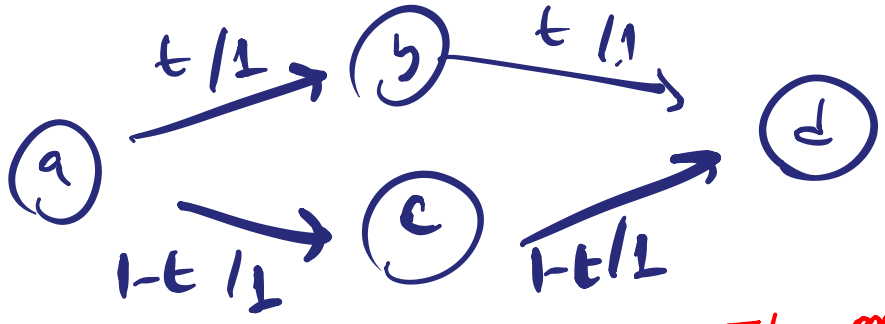
max flow = 1 + 4 = 5

min cut: {(4, 6), (3, 5), (1, 2)}

3. a. Does the maximum-flow problem always have a unique solution? Would your answer be different for networks with different capacities on all their edges?

b. Answer the same questions for the minimum-cut problem of finding a cut of the smallest capacity in a given network.

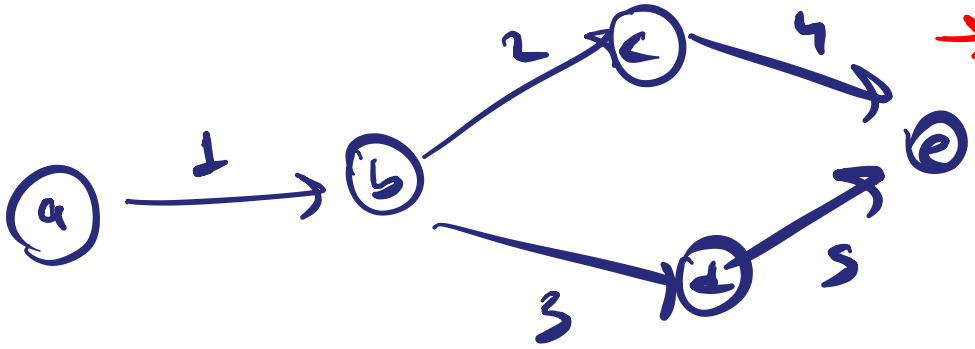
a)



If $t=0$ → $a-c-d$

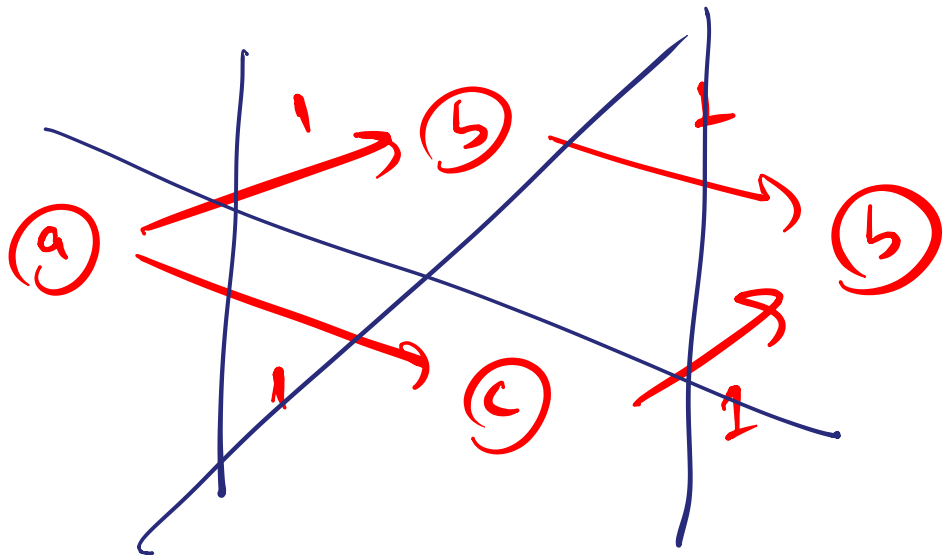
If $t=1$ → $a-b-d$

The maximum flow may have more than one optimal solution



→ for distinct edges, it may have more than one optimal solution

b)



{(1,2),(1,3)}

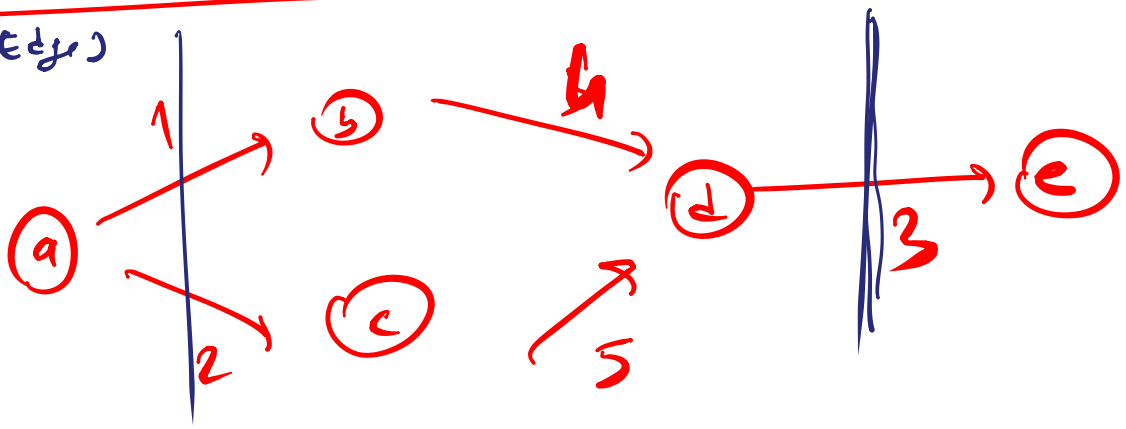
{(2,4)(3,4)}

{(1,2),(3,4)}

{(4,4),(1,3)}

minimum c-fs

Distinct Edges



{(a,b)(a,c)

{d,e}

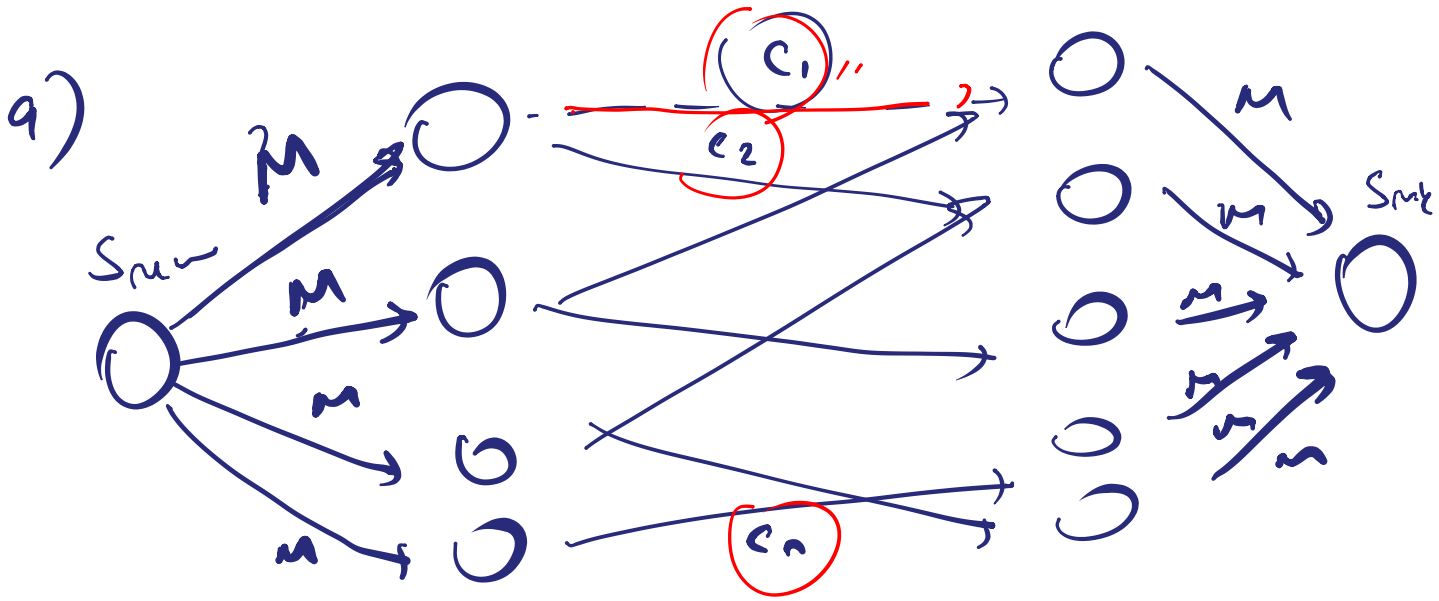
→ Not always unique.

value of Max flow = capacity of minimum cut

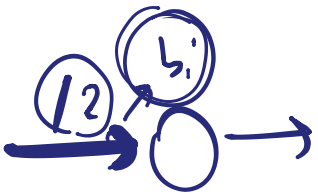
10.2

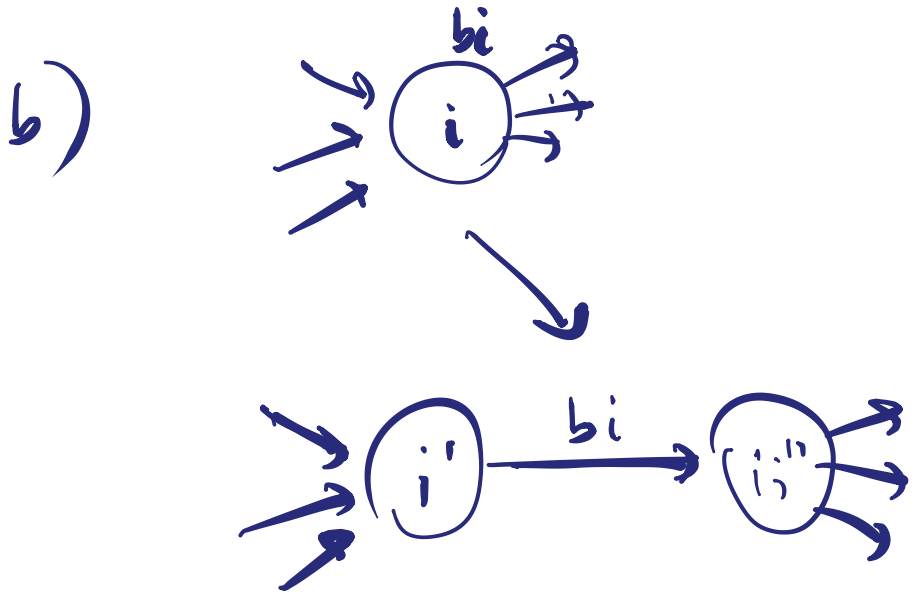
4. a. Explain how the maximum-flow problem for a network with several sources and sinks can be transformed into the same problem for a network with a single source and a single sink.

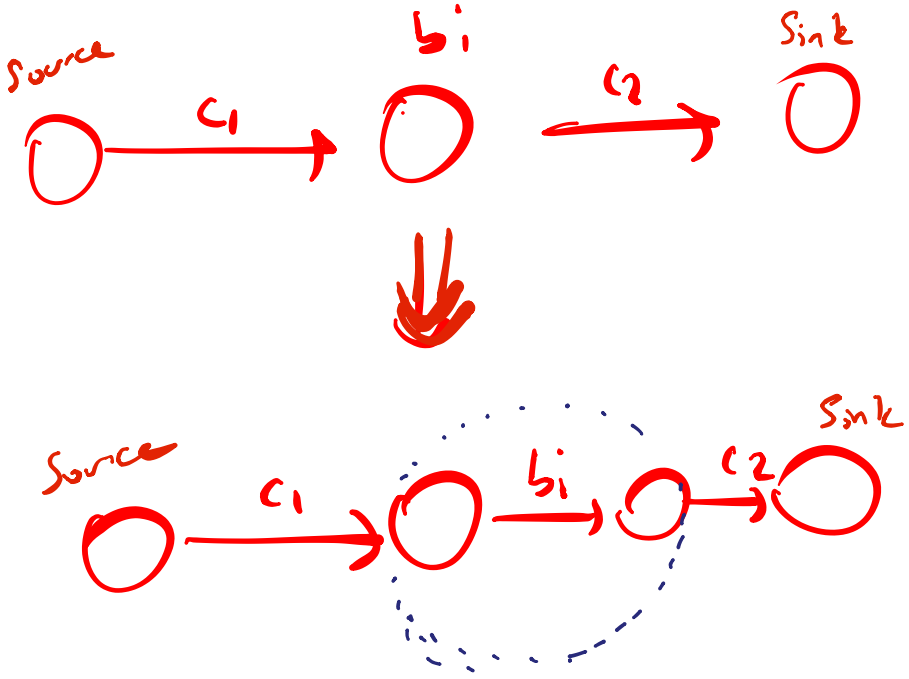
b. Some networks have capacity constraints on the flow amounts that can flow through their intermediate vertices. Explain how the maximum-flow problem for such a network can be transformed to the maximum-flow problem for a network with edge capacity constraints only.



$$M \geq \sum_{i=1}^{n} c_i$$







- All the edges entering and leaving node i in the original network should enter i' and i'' respectively.

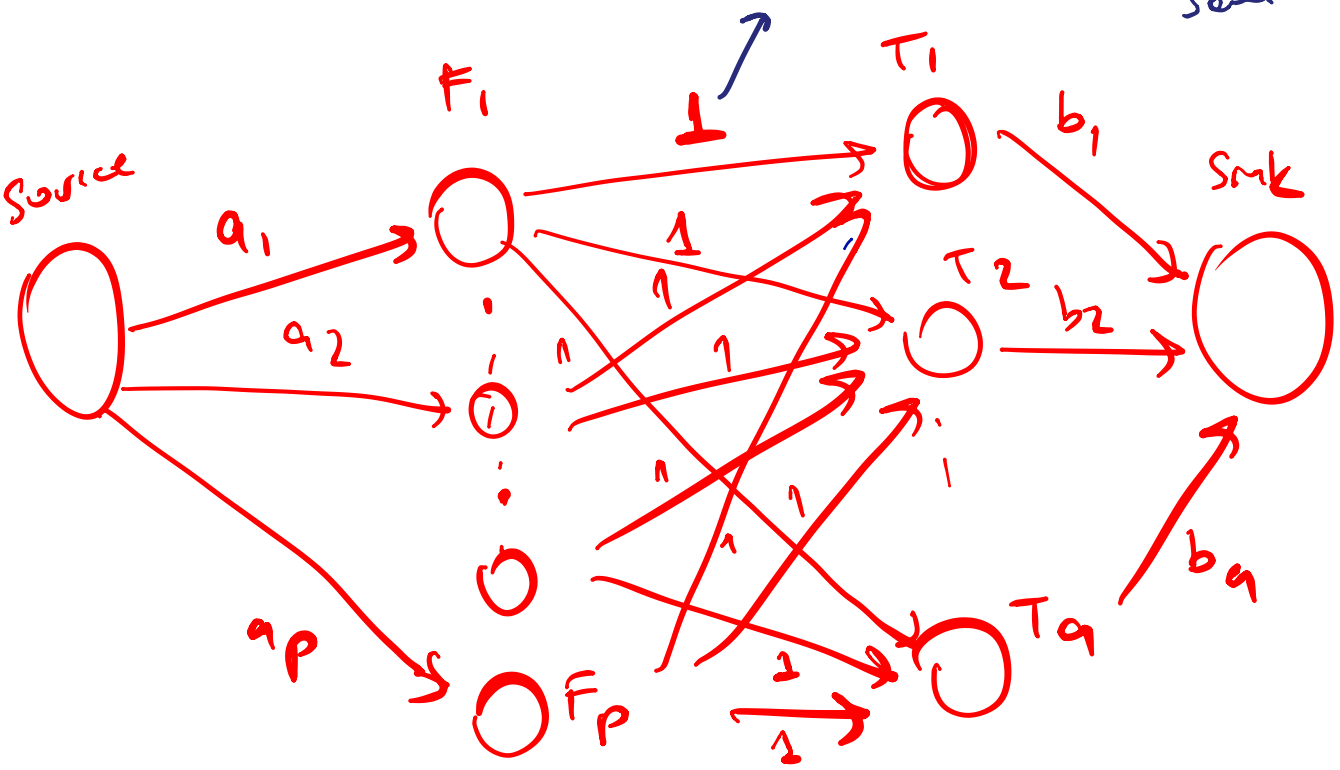
10.2

10. *Dining problem* Several families go out to dinner together. To increase their social interaction, they would like to sit at tables so that no two members of the same family are at the same table. Show how to find a seating arrangement that meets this objective (or prove that no such arrangement exists) by using a maximum-flow problem. Assume that the dinner contingent has p families and that the ith family has a_i members. Also assume that q tables are available and the jth table has a seating capacity of b_j. [Ahu93]

p families , Each family has a_i members
q Tables , each table has b_j capacity

$V = \{P_i, T_j\}$ $1 \le i \le p$, $1 \le j \le q$

Two members of the family are not at the same table



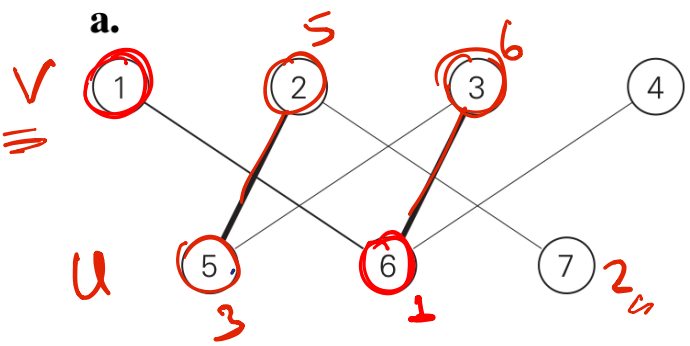
If maximum flow is equal to $\sum_{i=0}^{p} a_i$,

the problem has a solution, Otherwise no solution.

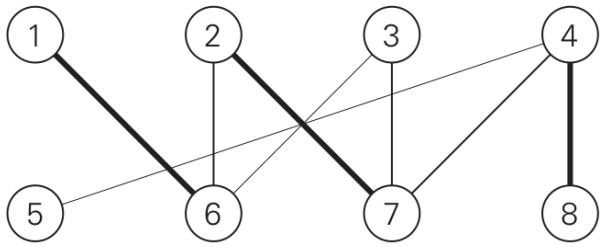
10.3

1. For each matching shown below in bold, find an augmentation or explain why no augmentation exists.

a.



b.



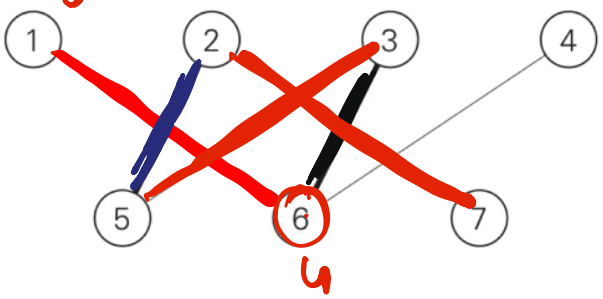
$M = \{(2,5), (3,6)\}$

Q: 1 4 6 3 5 2

→ 1 6 3 5 2 7 → Path

add add add

a.

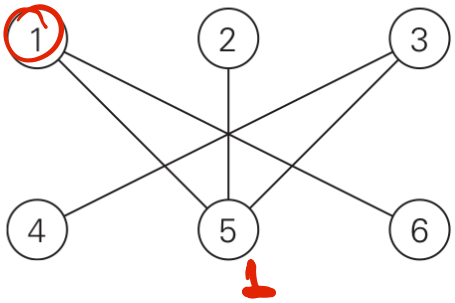


→ $M = \{(1,6), (3,5), (2,7)\}$

Q: 4 6 1 → No augmenting path

10.3

2. Apply the maximum-matching algorithm to the following bipartite graph:

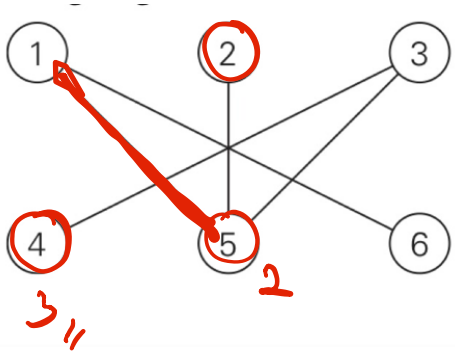


$M = \phi$

Q: 1 2 3

1 5 → Augment from 5.

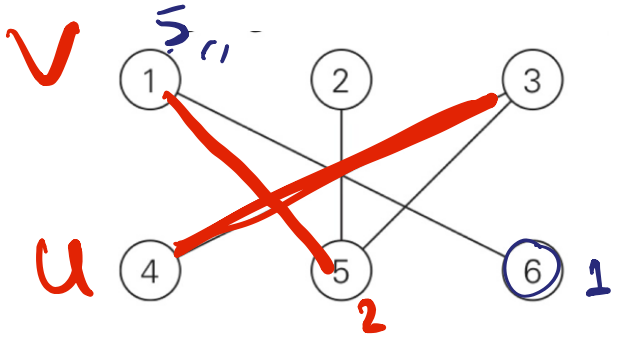
add



$M = \{(1,5)\}$

Q: 2 3 5

3 4 → Augment from 4

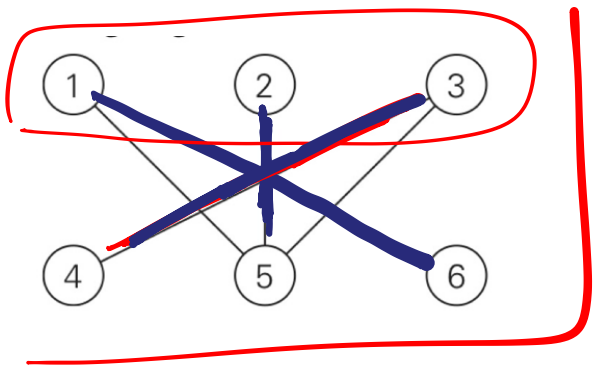


path: 2 5 1 6

add add

Q: 2 5 1 6

V



U

$M = \{(1,6), (2,5), (4,4)\}$

→ Perfect matching

Q: No

10.4

1. Consider an instance of the stable marriage problem given by the following ranking matrix:

M \ W	A	B	C
α	1, 3	2, 2	3, 1
β	3, 1	1, 3	2, 2
γ	2, 2	3, 1	1, 3

For each of its marriage matchings, indicate whether it is stable or not. For the unstable matchings, specify a blocking pair. For the stable matchings, indicate whether they are man-optimal, woman-optimal, or neither. (Assume that the Greek and Roman letters denote the men and women, respectively.)

3 2 1 = 6

M \ W	A	B	C
α	1, 3	2, 2	3, 1
β	3, 1	1, 3	2, 2
γ	2, 2	3, 1	1, 3

→ $(\alpha, A), (\beta, B), (\gamma, C)$

— stable? Yes

— Blocking pair? No, since each man has his best choice

— man-optimal

M \ W	A	B	C
α	1, 3	2, 2	3, 1
β	3, 1	1, 3	2, 2
γ	2, 2	3, 1	1, 3

(α, C), (β, A), (γ, B)

- Stable? Yes
  - No blocking pair, each woman has his best choice.
  - women optimal

M \ W	A	B	C
α	1, 3	2, 2	3, 1
β	3, 1	1, 3	2, 2
γ	2, 2	3, 1	1, 3

(α, B), (β, C), (γ, A)

γ offers → C, C not accept

- Stable? Yes
  - Neither men nor women optimal
  - No blocking pair

M \ W	A	B	C
α	1, 3	2, 2	3, 1
β	3, 1	1, 3	2, 2
γ	2, 2	3, 1	1, 3

(α, B), (β, A), (γ, C)

× α → A, → no

B → B → no

β → C → yes

stable ? No

Blocking pair (β, C)

M \ W	A	B	C
α	1, 3	2, 2	3, 1
β	3, 1	1, 3	2, 2
γ	2, 2	3, 1	1, 3

(α, A), (β, C), (γ, B)

→ α ×

→ β → B → No

→ γ → C → No

γ → A → yes

- (γ, A) → blocking pair

- unstable

M \ W	A	B	C
α	1, 3	2, 2	3, 1
β	3, 1	1, 3	2, 2
γ	2, 2	3, 1	1, 3

(α, C), (β, B), (γ, A)

→ α → A → No

α → B → yes

(α, B) → blocking pair

Unstable